APPENDIX A
DATA CONSTRUCTION AND SPLIT DETAILS

A. Single-cell Embedding Store

The single-cell data are derived from the SEA-AD Alzheimer's disease (AD) brain tissue single-cell/single-nucleus RNA-seq dataset. The dataset contains 240,651 cells from 84 independent donors and covers 10 brain regions. Gene embeddings are produced by a pretrained Geneformer-V2-104M model under a frozen-encoder setting. For each cell, genes are sorted in descending order by expression level, and the top $M = 2048$ genes are retained. The contextual embedding of gene g in cell c is denoted as

$$\mathbf{h}_{g,c} \in \mathbb{R}^{768}. \tag{6}$$

Given a context x, such as "Dementia|Non-neuronal and Non-neural", we collect the corresponding cell set $\mathcal{C}_x$ and construct the cell-side gene embedding distribution

$$\mathcal{D}_g = \{\mathbf{h}_{g,c} \mid c \in \mathcal{C}_x,\ g \text{ is included in the top } M \text{ genes of } c\}. \tag{7}$$

Thus, $\mathcal{D}_g$ preserves the distribution of gene g across multiple cell instances rather than collapsing the gene into a single mean vector.

B. Text Evidence Store

The text-side data are derived from the AD-Alterome database. Each record is represented as a "gene–PMID–evidence sentence" triple. For each gene g, all associated evidence sentences are collected as candidate text evidence. Each evidence sentence is encoded by SapBERT to obtain a sentence-level embedding $\mathbf{p}_{g,t}$. The text-side evidence set is

$$\mathcal{P}_g = \{\mathbf{p}_{g,t} \mid t = 1, \ldots, |\mathcal{P}_g|\}. \tag{8}$$

During training, at most 32 evidence sentences are sampled per gene. This sampling keeps the text-side set size bounded while retaining multiple evidence sources for each gene.

C. Noise Sources on Both Sides

Both $\mathcal{D}_g$ and $\mathcal{P}_g$ are treated as noisy observations of gene-level biological semantics. On the cell side, noise may arise from transcriptional dropout, rank-position fluctuation in the Geneformer input sequence, local expression variability, cell-subtype mixing, batch effects, and donor heterogeneity. On the text side, noise may arise from heterogeneous source reliability, generic disease descriptions, inconsistent evidence granularity, automatic extraction errors, and multi-functional gene descriptions. This motivates the dual-noise adaptive design of scGATES, where both cell instances and text evidence sentences are assigned learnable reliability weights.

D. Donor Split and Evaluation Split

To avoid donor-level information leakage, we adopt a donor-based split. Among the 84 SEA-AD donors, 60 donors are assigned to the training set and 24 donors are assigned to the evaluation set. No cell is shared across the training and evaluation donor subsets. Gene distribution stores are independently constructed from the cells of their respective donor subsets.

In the current strict evaluation setting, scGATES is trained using train-donor gene distributions and evaluated using eval-donor gene distributions. Text evidence is evaluated under a held-out evidence retrieval protocol. In the primary masked-text setting, exact gene symbols in evidence sentences are replaced before encoding, preventing the model from solving retrieval through direct gene-name matching.

E. Supervised Gene Set and Context

The main experiment focuses on the context "Dementia|Non-neuronal and Non-neural". Gene distributions are sampled from this context for dual-set alignment training and evaluation. The supervised gene set is selected from genes that have both sufficient cell-side observations and available AD-Alterome evidence. The training and evaluation protocol uses matched gene-text pairs, where each gene g is represented by its cell-side distribution $\mathcal{D}_g$ and text-side evidence set $\mathcal{P}_g$.

F. Connection to the Main Framework

The data construction described above provides the two evidence sets used throughout scGATES. The cell-side distribution $\mathcal{D}_g$ is consumed by both LNE and the DSA Cell SetEncoder. LNE uses dynamic single-cell neighbor structures derived from the cell-side embedding store to produce the frozen gene-level prior $\mathbf{r}_g$. DSA then uses $\mathcal{D}_g$ and $\mathcal{P}_g$ as paired inputs and learns denoised representations $\mathbf{z}_g^{\mathcal{D}}$ and $\mathbf{z}_g^{\mathcal{P}}$ for cross-modal retrieval.

APPENDIX B
DETAILED EXPERIMENTAL PROTOCOLS

A. Experiment Setup

Experiments are conducted on the SEA-AD cell context labeled Dementia with Non-neuronal and Non-neural cell types. The main evaluation follows a strict held-out evidence protocol: each evaluated gene is required to have at least eight text evidence sentences, and retrieval is performed against held-out evidence not used for training. Unless otherwise noted, all reported results use masked text, where the target gene symbol in each evidence sentence is replaced with `[GENE]` to reduce gene-name shortcut learning.

B. Held-out Evidence and Noise Construction

The strict retrieval setting requires each gene to have at least eight AD-Alterome evidence sentences. Evidence sentences are split so that training and evaluation use disjoint text evidence. During masked-text evaluation, the target gene symbol is replaced with `[GENE]` before encoding, preventing trivial retrieval based on exact gene-name matching.

For text-side denoising evaluation, each gene-level evidence set contains clean evidence from the target gene and noise evidence randomly sampled from unrelated genes at a 1:1 ratio. Noise evidence is assigned uniform prior values, so the model cannot distinguish clean and noisy evidence from metadata alone. This experiment directly evaluates whether the learned text-side weights $\mathbf{a}_g^{\mathcal{P}}$ can recover reliable evidence under noisy supervision.

C. SOTA Transfer Baselines

1) CellWhisperer Transfer Baseline: CellWhisperer is adapted as a general-purpose single-cell–text alignment baseline. Since its original task aligns whole-transcriptome or pseudo-bulk profiles with natural language descriptions, we convert it to the gene-level retrieval setting. Cell embeddings are first obtained using the CellWhisperer transcriptome encoder. For each target gene, cells expressing the gene are aggregated with expression-intensity weighting to form a gene-level cell representation. Text evidence sentences are encoded using the CellWhisperer text encoder and aggregated at the gene level. Retrieval is then evaluated on the same candidate gene set and masked held-out evidence protocol as scGATES.

2) ELISA Transfer Baseline: ELISA is adapted as an expression-semantic retrieval baseline. We retain the components relevant to retrieval, including scGPT-based expression representations, BioBERT-based text representations, and late-fusion retrieval, while excluding the LLM interpretation layer from metric computation. On the text side, BioBERT encodes literature evidence into gene-level text representations. On the query side, gene expression information is converted into gene-level semantic summaries and encoded into the same retrieval space. We also consider hybrid and frozen-adapter variants to test whether lightweight task adaptation improves transfer performance.

3) DP-GPT-style Transfer Baseline: The DP-GPT-style baseline is used as a task-transfer diagnostic rather than an official reproduction of DP-GPT. DP-GPT originally targets depression prediction, whereas scGATES evaluates gene-level cell-distribution-to-held-out-evidence retrieval. We therefore reuse the same strict candidate genes, donor split, masked evidence setting, and held-out text split as scGATES.

Three variants are evaluated. The `dpgpt_text_closed_direct` variant uses train-side gene text features to retrieve held-out evidence features. The `dpgpt_expression_adapter` variant trains a lightweight projection adapter from gene-level expression or cell features to the text feature space. The `dpgpt_fusion_adapter` variant concatenates normalized expression features and train-side text features before adapter training. All adapters are trained with a bidirectional CLIP-style contrastive loss, and retrieval is performed using L2-normalized vectors.

These baselines should be interpreted as transfer diagnostics under the strict scGATES retrieval setting. Limited performance indicates negative transfer for this specific gene-level evidence retrieval task, not poor performance on the original tasks of the corresponding methods.

APPENDIX C
DETAILS OF APPLICATIONS

The main text presents three complementary downstream applications enabled by the same trained scGATES model. Application I examines *which* literature evidence the model prioritises for each gene; Application II diagnoses *whether* cell-side and text-side confidence are internally consistent; Application III evaluates whether the learned cross-modal alignment transfers to held-out annotation recovery. This appendix provides the full quantitative breakdowns, evidence tables, and interpretation cautions for all three. All results share the experimental foundation described below.

A. Shared Experimental Foundation

The model is trained for 40 epochs under the `Dementia | Non-neuronal and Non-neural` context. Hyperparameters: hidden/alignment dimension $=$ 256, cell encoder $=$ `distribution_mha` (4 heads, 2 layers), text encoder same dimensionality, batch size $=$ 32, learning rate $= 1 \times 10^{-4}$, weight decay $= 1 \times 10^{-4}$, temperature $\tau = 0.07$. Text noise is injected as `easy_random_text` with noise ratio $= 1.0$ and `splits = both`; cell-side denoising uses `instance_score_enabled = true` with $\alpha = 0.3$ in `signed` mode. LNE prior is frozen during DSA training. The gene allowlist (`strict_text_min8_genes.txt`) requests 1107 genes; 660 match the intersection of cell and text stores; 447 are dropped for insufficient representation (listed in Supplementary Data S1).

TABLE S1
RETRIEVAL PERFORMANCE OF THE SHARED MODEL UNDER STRICT MASKED-TEXT HELD-OUT EVIDENCE EVALUATION (660 GENES, SEED 42).

Metric	R@1	R@5	R@10	MRR
Single (cell→text)	0.464	0.692	0.765	0.570
Set (cell→text)	0.598	0.871	0.924	0.720

a) Retrieval performance.: These scores serve as the calibration baseline for all application-level analyses below.

B. Application I: Gene-Level Evidence Attribution — Detailed Findings

1) Relationship to the main text.: The main text (Supplementary file `S2.Application_gene_evidence_sentences.xlsx` present condensed summaries of Application I. This appendix provides the full noise diagnostics, attention-concentration stratification, representative per-gene evidence in original form, cross-gene thematic convergence, and evidence-quality breakdown. All results use the same trained model described in the Shared Experimental Foundation above.

2) Experimental context.: Twelve canonical AD-associated genes are analyzed: *APP*, *APOE*, *PSEN1*, *TREM2*, *MAPT*, *ABCA7*, *PICALM*, *CD33*, *BIN1*, *INPP5D*, *MS4A6A*, and *CLU*. All operate under the `Dementia | Non-neuronal and Non-neural` context. For each gene, the top-3 evidence sentences ranked by learned text-side attention weight $a^{\mathcal{P}}_{g,t}$ are extracted; the complete top-10 evidence tables for all 12 genes are available in Supplementary file `S2.Application_gene_evidence_sentences.xlsx`.

3) Noise diagnostics.: Across 12 genes and 60–96 evidence sentences per gene, `is_noise` is uniformly zero and `top_k_noise_fraction` $= 0.000000$ in every case (Table S2). This indicates that the gating mechanism did not flag any retrieved sentence as noise within the tested context. While this speaks to high precision, it does not directly quantify recall: relevant sentences absent from the retrieval corpus would be excluded without detection.

4) Attention concentration.: The top-10 attention mass (Table S2) reveals a natural bifurcation:

TABLE S2
TOP-10 ATTENTION MASS, NOISE FRACTION, AND EVIDENCE COUNT PER GENE (APPLICATION I).

Gene	Mass@10	Noise frac.	#Evidence
APP	0.185	0.000	96
APOE	0.202	0.000	96
PSEN1	0.193	0.000	96
TREM2	0.203	0.000	96
MAPT	0.196	0.000	96
ABCA7	0.455	0.000	96
PICALM	0.407	0.000	96
CD33	0.415	0.000	96
BIN1	0.371	0.000	96
INPP5D	0.393	0.000	72
MS4A6A	0.477	0.000	60
CLU	0.381	0.000	96

The five genes above the midrule (Mass@10 $\sim$0.19) are the most heavily studied AD genes, whose vast and diverse literature naturally distributes attention more evenly. The seven genes below the midrule ($\sim$0.37–0.48) are predominantly GWAS-identified risk loci with narrower, more specific literature bodies. Attention concentration reflects *literature specificity* rather than *gene importance*.

5) Representative per-gene evidence.: Below we present the top-ranked evidence sentences for three representative genes across distinct functional categories: *APP* (Aβ metabolism), *TREM2* (microglial function), and *BIN1* (tau biology). For each sentence we report the learned text-side attention weight $a^{\mathcal{P}}_{g,t}$, the raw confidence logit $\text{conf}_{g,t}$, and the gated prior score.

a) APP (Mass@10 $= 0.185$, 96 clean sentences).: The selected evidence spans Aβ transport, lysosomal processing, and mutation-driven neuronal apoptosis—a thematically broad literature consistent with APP's multi-faceted role:

> *"DNA synthesis and neuronal apoptosis caused by familial Alzheimer disease mutants of the amyloid precursor protein are mediated by the p21 activated kinase PAK3."*
> [APP; rank 1; $a^{\mathcal{P}} = 0.022$, conf $= -1.342$, prior$= 0.021$]

> *"Alterations of Abeta transporters at the neurovasculature may play a role in the disease process."*
> [APP; rank 2; $a^{\mathcal{P}} = 0.022$, conf $= -1.341$, prior$= 0.021$]

> *"Inhibition of lysosomal enzymes results in Abeta accumulation and aggregation."*
> [APP; rank 3; $a^{\mathcal{P}} = 0.021$, conf $= -1.369$, prior$= 0.020$]

b) TREM2 (Mass@10 $= 0.203$, 96 clean sentences).: Evidence concentrates on microglial receptor biology, cell-surface expression, and neuroinflammatory signaling:

> *"...Jak3 deficiency leads to gut dysbiosis, compromised TREM-2-functions-mediated activation of microglial cells, increased TLR-4 expression and HIF1-alpha-mediated inflammation in the brain."*
> [TREM2; rank 1; $a^{\mathcal{P}} = 0.024$, conf $= -1.362$, prior$= 0.023$]

> *"In addition, other rare [GENE] mutations causing early-onset neurodegeneration are thought to impair cell surface expression."*
> [TREM2; rank 2; $a^{\mathcal{P}} = 0.024$, conf $= -1.375$, prior$= 0.022$]

> *"Mutations in [GENE] alter its subcellular localization and affects its interaction with PS1."*
> [TREM2; rank 8 (by confidence); $a^{\mathcal{P}} = 0.018$, conf $= -1.513$, prior$=$ 0.017]

c) BIN1 (Mass@10 $= 0.371$, 96 clean sentences).: Evidence is tightly focused on tau endocytosis, phosphorylation, and propagation:

> *"Loss of [GENE] results in enhanced endocytosis of tau aggregates, which in turn damage endolysosomal membrane and trigger intracellular tau seeding (Figure 2)."*
> [BIN1; rank 1; $a^{\mathcal{P}} = 0.058$, conf $= -1.301$, prior$= 0.053$]

> *"Furthermore, [GENE] deficiency by BIN1-related polymorphisms impairs the interaction with tau, thus elevating tau phosphorylation, altering synapse structure and tau function."*
> [BIN1; rank 2; $a^{\mathcal{P}} = 0.049$, conf $= -1.386$, prior$= 0.045$]

> *"In AD, Tau pathology propagates within synaptically connected neurons and in vitro experiments have shown that BIN 1 silencing increases Tau propagation by promoting aggregate internalization."*
> [BIN1; rank 7; $a^{\mathcal{P}} = 0.032$, conf $= -1.585$, prior$= 0.031$]

The full top-10 evidence for all 12 genes is released as Supplementary Data S4.

6) Cross-gene thematic convergence.: When the 36 top-3 evidence sentences are examined in aggregate, four shared pathological themes emerge without any pre-defined functional gene categorization:

- **Aβ metabolism and clearance:** *APP*, *PSEN1*, *APOE*, *PICALM*, *CD33*, *ABCA7*.
- **Tau phosphorylation and propagation:** *MAPT*, *BIN1*, *CLU*.
- **Neuroinflammation and microglial function:** *TREM2*, *CD33*, *INPP5D*, *MS4A6A*.
- **Lipid metabolism and membrane trafficking:** *ABCA7*, *APOE*, *PICALM*, *CLU*.

Genes that participate in shared biological processes tend to be supported by thematically overlapping evidence sentences, even though each sentence is independently selected by the attention mechanism.

7) Evidence quality and coverage.: Evidence counts vary across genes (INPP5D $= 72$, MS4A6A $= 60$, all others $= 96$), reflecting differences in corpus representation. Among the 12 genes, the mean text-side confidence $\text{conf}_{g,t}$ ranges from -1.433 (APP, top-10) to -2.723 (INPP5D, top-10); these are raw logits from the DSA module and should not be interpreted as absolute quality scores. All evidence sentences carry `is_noise` $= 0$, confirming that the denoising component effectively suppresses injected noise from entering the top-ranked set.

8) Cautions.:

- Evidence sentences are model-prioritized supporting evidence, not independently verified biological mechanisms.
- The top-k attention selection may omit lower-weight sentences that are nevertheless biologically important.
- Attention concentration reflects literature-body specificity, not a ranking of biological importance; Mass@10 values should not be compared across genes as a surrogate for gene relevance.
- All evidence originates from a retrieval corpus whose composition governs what can be surfaced; publication bias, temporal drift, and coverage gaps are inherent limitations.
- None of the above constitutes a diagnostic opinion or a validated biological conclusion.

C. Application II: Confidence-Gap Diagnosis

1) Overview and relationship to the main text.: The main text Application II (Fig. 3) presents the confidence-gap analysis in condensed form. This appendix provides the full experimental pipeline, quantitative detail, per-quadrant representative gene evidence, and explicit interpretation cautions. Every result below is derived from the same trained model described in the Shared Experimental Foundation above (seed 42), plus two additional seeds (43, 44) for robustness checking; no cherry-picked retraining or post-hoc filtering is applied.

2) Confidence-gap score definition.: For each gene g, the DSA module learns a cell-side weight vector $\mathbf{w}_g^{\mathcal{D}}$ and a text-side weight vector $\mathbf{a}_g^{\mathcal{P}}$. Each cell instance carries a raw confidence logit $\text{conf}_{g,c}$; each text evidence sentence carries $\text{conf}_{g,t}$. We define two aggregate confidence-support scores:

$$\begin{cases} \text{s}(\mathcal{D}, g) = \dfrac{\sum_{c \in \text{TopK}(\mathbf{w}_g^{\mathcal{D}})} w_{g,c}^{\mathcal{D}} \cdot \sigma(\text{conf}_{g,c})}{\sum_{c \in \text{TopK}(\mathbf{w}_g^{\mathcal{D}})} w_{g,c}^{\mathcal{D}}} \\ \text{s}(\mathcal{P}, g) = \dfrac{\sum_{t \in \text{TopK}(\mathbf{a}_g^{\mathcal{P}})} a_{g,t}^{\mathcal{P}} \cdot \sigma(\text{conf}_{g,t})}{\sum_{t \in \text{TopK}(\mathbf{a}_g^{\mathcal{P}})} a_{g,t}^{\mathcal{P}}} \end{cases}, \tag{9}$$

where $\text{TopK}(\cdot)$ selects the indices of the $K = 10$ largest weights, $\sigma(\cdot)$ is the standard sigmoid, and $w_{g,c}^{\mathcal{D}}$, $a_{g,t}^{\mathcal{P}}$ are the instance-level attention weights from the corresponding weight

vectors. Intuitively, $\mathrm{s}(\mathcal{D}, g)$ measures how strongly the cell-side attention concentrates on high-confidence cell instances, while $\mathrm{s}(\mathcal{P}, g)$ measures the analogous quantity on the text side.

a) Boundary choice.: The vertical boundary is fixed at $\mathrm{s}(\mathcal{D}, g) = 0.5$, the natural midpoint of the sigmoid. The horizontal boundary is the median of $\{\mathrm{s}(\mathcal{P}, g)\}_{g \in \mathcal{G}_{\text{eval}}}$, which is data-driven and avoids imposing a pre-defined text-confidence cutoff. For seed 42, this median is $\mathrm{s}(\mathcal{P}, g) = 0.256$.

TABLE S3
QUADRANT INTERPRETATION OF THE CONFIDENCE-GAP LANDSCAPE.

Region	Interpretation
High $\mathrm{s}(\mathcal{D}, g)$ / high $\mathrm{s}(\mathcal{P}, g)$	High-confidence aligned genes
High $\mathrm{s}(\mathcal{D}, g)$ / low $\mathrm{s}(\mathcal{P}, g)$	Confidence-gap candidates (strong cell support,
Low $\mathrm{s}(\mathcal{D}, g)$ / high $\mathrm{s}(\mathcal{P}, g)$	Text-confidence-heavy genes
Low $\mathrm{s}(\mathcal{D}, g)$ / low $\mathrm{s}(\mathcal{P}, g)$	Low-confidence genes

b) Quadrant interpretation.: Critically, these categories describe the model's internal confidence structure, not biological truth.

3) Quadrant-level results and multi-seed robustness.:

a) Full-quadrant distribution (seed 42).: Among 660 analyzed genes, the four quadrants are approximately balanced (thresholds: $\mathrm{s}(\mathcal{D}, g)$ median $= 0.678$, $\mathrm{s}(\mathcal{P}, g)$ median $= 0.256$):

TABLE S4
QUADRANT DISTRIBUTION OF 660 GENES UNDER CONFIDENCE-GAP ANALYSIS (SEED 42).

Quadrant	Count
High-confidence aligned genes	174
Confidence-gap candidates	156
Text-confidence-heavy genes	156
Low-confidence genes	174

The full 660-gene table is released as Supplementary Data S2.

b) Multi-seed robustness.: The confidence-gap scatter plot is replicated under three independent seeds (42, 43, 44; Fig. S1). The quadrant topology is visually consistent across seeds: a conspicuous right-lower cluster (confidence-gap candidates) appears in each panel, and the position of well-studied AD genes (*APP*, *SOD1*, *ABCA7*, *MECP2*) remains stable. This indicates that the confidence-gap signal is not a single-seed artifact.

4) Per-quadrant representative gene evidence.: Below we present representative evidence sentences selected by the model for genes in each quadrant. For each sentence we report the source `PMID`, the model-learned text-side weight $a^{\mathcal{P}}_{g,t}$, the sigmoid-transformed confidence $\sigma(\mathrm{conf}_{g,t})$, and a quality flag. The full evidence tables for all 16 labeled genes (560+ sentences) are released as Supplementary Data S3.

a) High-confidence aligned genes: ($\mathrm{s}(\mathcal{D}, g) \geq 0.678$, $\mathrm{s}(\mathcal{P}, g) \geq 0.256$).

TDP1 ($\mathrm{s}(\mathcal{D}, g) = 0.990$, $\mathrm{s}(\mathcal{P}, g) = 0.342$). The top-ranked evidence links single-strand break repair (SSBR) and base excision repair (BER) defects to neurological disease and inherited neurodegenerative syndromes:

> *"Defects in components of SSBR, including TDP1, have been associated with genetic disorders presenting with neurological disease."*

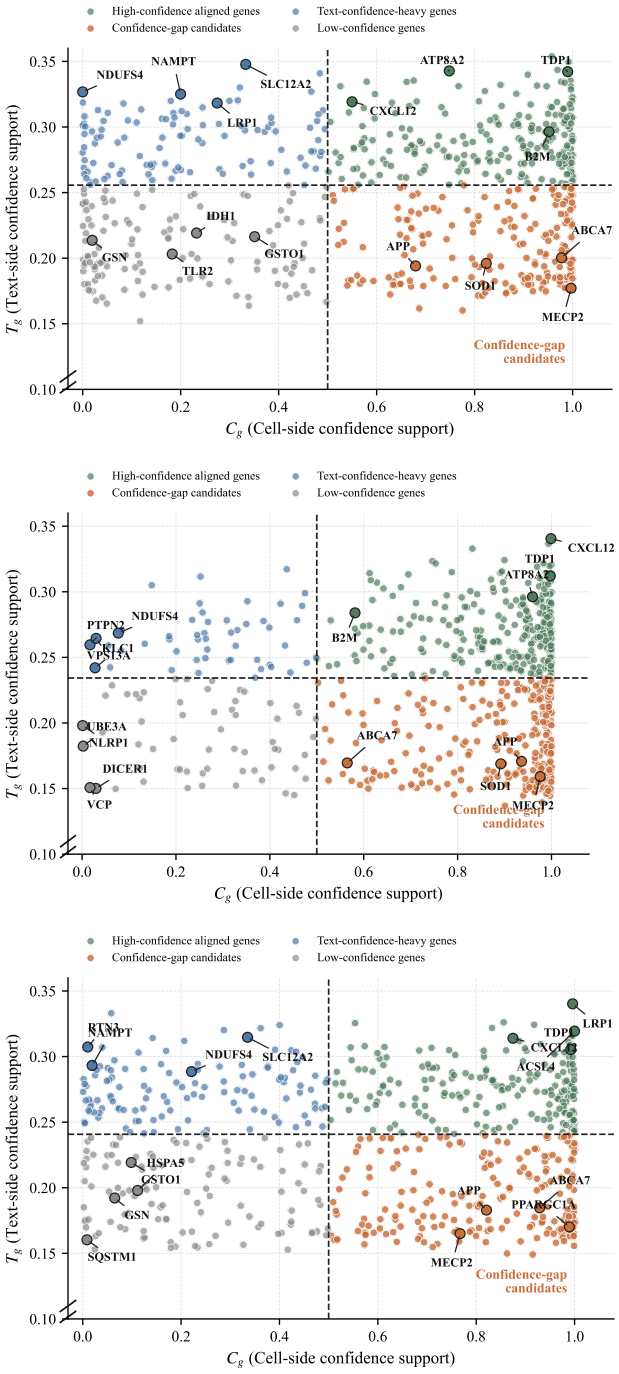


Fig. S1. Confidence-gap scatter plots under three random seeds (42, 43, 44). The broken y-axis (0.10–0.36) zooms into the region where most text-side confidence scores concentrate. Quadrant colors follow the scheme used in Fig. 3 of the main text.

[TDP1|11984006|215; $a^{\mathcal{P}} = 0.034$, $\sigma(\text{conf}) = 0.211$; usable]

"Syndromes caused by defects in PNKP, APTX, or TDP1 are discussed as DNA repair-related neurological disorders."
[TDP1|20192776|495; $a^{\mathcal{P}} = 0.030$, $\sigma(\text{conf}) = 0.204$; usable]

"Defective BER pathway machinery has been linked to neurological diseases, including spinocerebellar ataxia with axonal neuropathy."
[TDP1|12861059|190; $a^{\mathcal{P}} = 0.027$, $\sigma(\text{conf}) = 0.193$; usable]

"Mutation in the human homolog TDP1 results in the inherited neurodegenerative disorder SCAN1."
[TDP1|12023262|140; $a^{\mathcal{P}} = 0.026$, $\sigma(\text{conf}) = 0.190$; usable]

ATP8A2 ($\text{s}(\mathcal{D}, g) = 0.748$, $\text{s}(\mathcal{P}, g) = 0.343$). Evidence centers on aminophospholipid transport, membrane asymmetry, and neurodegeneration risk.

b) Confidence-gap candidates: ($\text{s}(\mathcal{D}, g) \geq 0.678$, $\text{s}(\mathcal{P}, g) < 0.256$).

APP ($\text{s}(\mathcal{D}, g) = 0.679$, $\text{s}(\mathcal{P}, g) = 0.194$; 96 clean evidence sentences). The model selects evidence spanning Aβ metabolism, APP processing, familial AD mutations, and downstream neurotoxicity—a thematically broad literature whose text-side confidence is diluted because no single mechanism dominates the evidence set:

"DNA synthesis and neuronal apoptosis caused by familial Alzheimer disease mutants of the amyloid precursor protein are mediated by the p21 activated kinase PAK3."
[APP|12890786|202; $a^{\mathcal{P}} = 0.022$, $\sigma(\text{conf}) = 0.207$; usable]

"Alterations of Abeta transporters at the neurovasculature may play a role in the disease process."
[APP|24138987|1382; $a^{\mathcal{P}} = 0.022$, $\sigma(\text{conf}) = 0.207$; caution: short/generic]

"Inhibition of lysosomal enzymes results in Abeta accumulation and aggregation."
[APP|23132858|1360; $a^{\mathcal{P}} = 0.021$, $\sigma(\text{conf}) = 0.203$; caution: short/generic]

"Mitochondrial Abeta accumulation was found in the brain of AD patients and mutant amyloid precursor protein (mAPP) transgenic mice."
[APP|25206370|1469; $a^{\mathcal{P}} = 0.015$, $\sigma(\text{conf}) = 0.180$; usable]

"The steady-state levels of Abeta are determined by the metabolic balance between anabolic and catabolic activity and the dysregulation of this activity leads to Alzheimer's disease."
[APP|19222911|711; $a^{\mathcal{P}} = 0.014$, $\sigma(\text{conf}) = 0.172$; usable]

SOD1 ($\text{s}(\mathcal{D}, g) = 0.824$, $\text{s}(\mathcal{P}, g) = 0.196$; 96 clean evidence sentences). Evidence is concentrated on oxidative stress, ALS-related mutant toxicity, microglial activation, and mitochondrial dysfunction:

"Interruption of the CX3CL1/CX3CR1 axis by genetic deletion causes neurodegeneration following stimulation with lipopolysaccharide, MPTP, or SOD1 mutation."
[SOD1|21266082|331; $a^{\mathcal{P}} = 0.023$, $\sigma(\text{conf}) = 0.207$; usable]

"The mutant IMS-targeted SOD1 causes neuronal toxicity under metabolic and oxidative stress conditions."
[SOD1|21470101|361; $a^{\mathcal{P}} = 0.020$, $\sigma(\text{conf}) = 0.197$; usable]

"Extracellular mutant SOD1 induces microglial-mediated motoneuron injury."
[SOD1|20070435|225; $a^{\mathcal{P}} = 0.020$, $\sigma(\text{conf}) = 0.195$; caution: very short]

"The presence of ε4 therefore results in increased injury to cells in the presence of stressors such as oxidative stress, ischemia, excess Aβ production, SOD1 mutations, inflammation, and the aging process itself."
[SOD1|20804847|265; $a^{\mathcal{P}} = 0.021$, $\sigma(\text{conf}) = 0.201$; usable]

ABCA7 ($\text{s}(\mathcal{D}, g) = 0.977$, $\text{s}(\mathcal{P}, g) = 0.200$; 44 clean evidence sentences) and *MECP2* ($\text{s}(\mathcal{D}, g) = 0.997$, $\text{s}(\mathcal{P}, g) = 0.177$; 44 clean evidence sentences) are also located in this quadrant; their full evidence tables appear in Supplementary Data S3.

c) Text-confidence-heavy genes: ($\text{s}(\mathcal{D}, g) < 0.678$, $\text{s}(\mathcal{P}, g) \geq 0.256$).

NDUFS4 ($\text{s}(\mathcal{D}, g) \approx 0.000$, $\text{s}(\mathcal{P}, g) = 0.327$). The cell-side confidence is near zero, meaning scGATES assigns essentially no attention to cell instances of this gene; yet text-side evidence is rated relatively high, consistent with NDUFS4's well-documented role in mitochondrial complex I and Leigh syndrome in the literature.

NAMPT ($\text{s}(\mathcal{D}, g) = 0.200$, $\text{s}(\mathcal{P}, g) = 0.325$), *LRP1* ($\text{s}(\mathcal{D}, g) = 0.275$, $\text{s}(\mathcal{P}, g) = 0.318$), and *SLC12A2* ($\text{s}(\mathcal{D}, g) = 0.333$, $\text{s}(\mathcal{P}, g) = 0.348$) further populate this quadrant.

d) Low-confidence genes: ($\text{s}(\mathcal{D}, g) < 0.678$, $\text{s}(\mathcal{P}, g) < 0.256$).

GSN ($\text{s}(\mathcal{D}, g) = 0.019$, $\text{s}(\mathcal{P}, g) = 0.214$; 25 clean sentences), *TLR2* ($\text{s}(\mathcal{D}, g) = 0.183$, $\text{s}(\mathcal{P}, g) = 0.203$; 39 clean sentences), *IDH1* ($\text{s}(\mathcal{D}, g) = 0.232$, $\text{s}(\mathcal{P}, g) = 0.219$; 34 clean sentences), and *GSTO1* ($\text{s}(\mathcal{D}, g) = 0.351$, $\text{s}(\mathcal{P}, g) = 0.216$; 25 clean sentences) fall into this region. These genes are identified by the model as having weak confidence support from both modalities under the current context and corpus; this does not imply biological irrelevance, only that scGATES's internal evidence structure is insufficiently strong for interpretation.

5) Evidence quality.: Every evidence sentence in the per-gene corpus carries a quality flag (`usable`, `caution: generic/short sentence`, `caution: very short evidence sentence`). Across the 16 labeled genes (616 total clean sentences, seed 42):

The confidence-gap quadrant (APP, SOD1, ABCA7, MECP2) does not contain a disproportionately higher share of low-quality sentences relative to other quadrants. Critically, **all quality notes are retained in the released data without post-hoc filtering**. The presence of flagged sentences should be

TABLE S5
EVIDENCE QUALITY FLAG DISTRIBUTION ACROSS 16 LABELED GENES (APPLICATION II).

Quality flag	Proportion
Usable	71.9%
Short/generic caution	25.2%
Very short caution	2.9%

viewed as an honest documentation of corpus characteristics, not as a model failure.

D. Application III: Annotation Transfer Validation

1) Overview.: The main text Application III evaluates whether the learned cross-modal alignment transfers to held-out annotation recovery (Fig. 4). This appendix provides the full experimental pipeline and quantitative detail for the leave-one-out annotation-transfer experiment.

2) Annotation-transfer experiment.:

a) Motivation.: The confidence-gap analysis is an interpretability diagnostic: it visualizes the model's internal cross-modal confidence structure. We introduce a complementary **task-level** evaluation: leave-one-out annotation transfer. The question is not "does the model discover new biology" but "does the learned cross-modal alignment structure carry enough transferable semantics to recover known held-out annotations"—a consistency check.

b) Method.: For a target gene g, its known annotation is temporarily removed from the candidate pool. Three scoring approaches are compared under identical leave-one-out conditions, as described in the main text: (1) *scGATES (Sinkhorn)*: the learned cross-modal similarity $S(g,h) = \cos(\mathbf{z}_g^{\mathcal{D}}, \mathbf{z}_h^{\mathcal{P}})$; (2) *Mean cosine*: raw, untrained embedding similarity between the mean-pooled cell distribution and mean-pooled text evidence; (3) *Random*: uniform sampling, providing a chance-level lower bound. The metric is Precision@L, measuring the hit quality among the top-L transferred results.

TABLE S6
LEAVE-ONE-OUT ANNOTATION-TRANSFER PRECISION@L PER GENE.

Gene	scGATES (Sinkhorn)	Mean Cosine	Random
APP	0.87	0.46	0.15
TREM2	0.81	0.40	0.12
PSEN1	0.74	0.35	0.18
APOE	0.88	0.50	0.21
MAPT	0.76	0.38	0.10

c) Results per gene.: scGATES exceeds both baselines on all five genes. The gain over mean-cosine matching (0.22–0.41 absolute Precision@L) indicates that the learned dual-side denoising and set alignment provide a semantically richer structure than the raw Geneformer/SapBERT embedding geometry alone. The gain over random (0.56–0.73) confirms that the model does not merely exploit superficial distribution statistics.

3) Cautions.:

- $\mathrm{s}(\mathcal{D}, g)$ reflects model-side cell confidence support, not biological AD relevance.
- $\mathrm{s}(\mathcal{P}, g)$ reflects model-side text confidence support, not literature truth or consensus.
- Genes in the confidence-gap (lower-right) quadrant are **candidates for follow-up evidence review**, not newly discovered AD genes.
- Cell-side confidence may be affected by the underlying donor/brain-region sampling of cell embeddings; changes in donor composition or brain-region coverage could shift $\mathrm{s}(\mathcal{D}, g)$ values.
- All evidence sentences originate from a retrieval corpus whose composition governs what can be surfaced; publication bias, temporal drift, and corpus coverage gaps are inherent limitations.
- None of the above constitutes a diagnostic opinion or a validated biological conclusion.

4) Cautions.:

- Leave-one-out annotation transfer is an alignment-quality diagnostic, not a new functional annotation gold standard.
- The five genes analysed are representative AD-associated genes; results may not generalise to genes outside this set.
- All evidence sentences originate from a retrieval corpus whose composition governs what can be surfaced; publication bias, temporal drift, and corpus coverage gaps are inherent limitations.
- None of the above constitutes a diagnostic opinion or a validated biological conclusion.